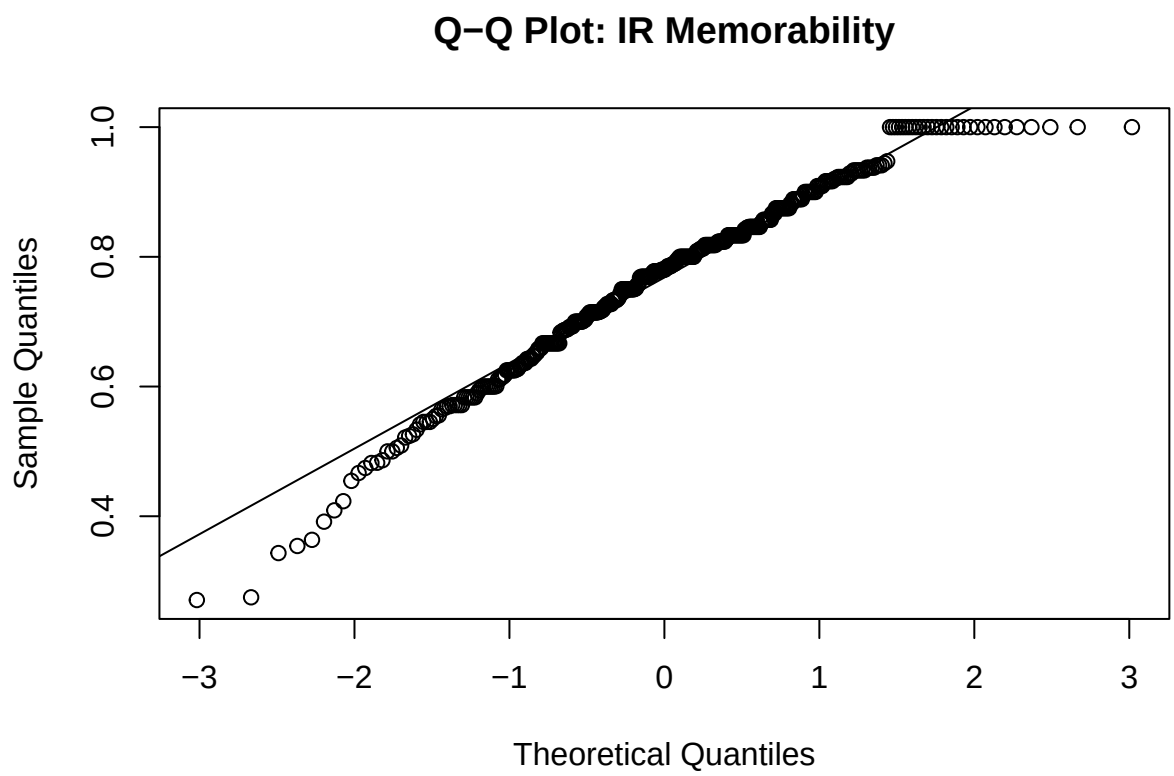
Sentence Memorability Graphs

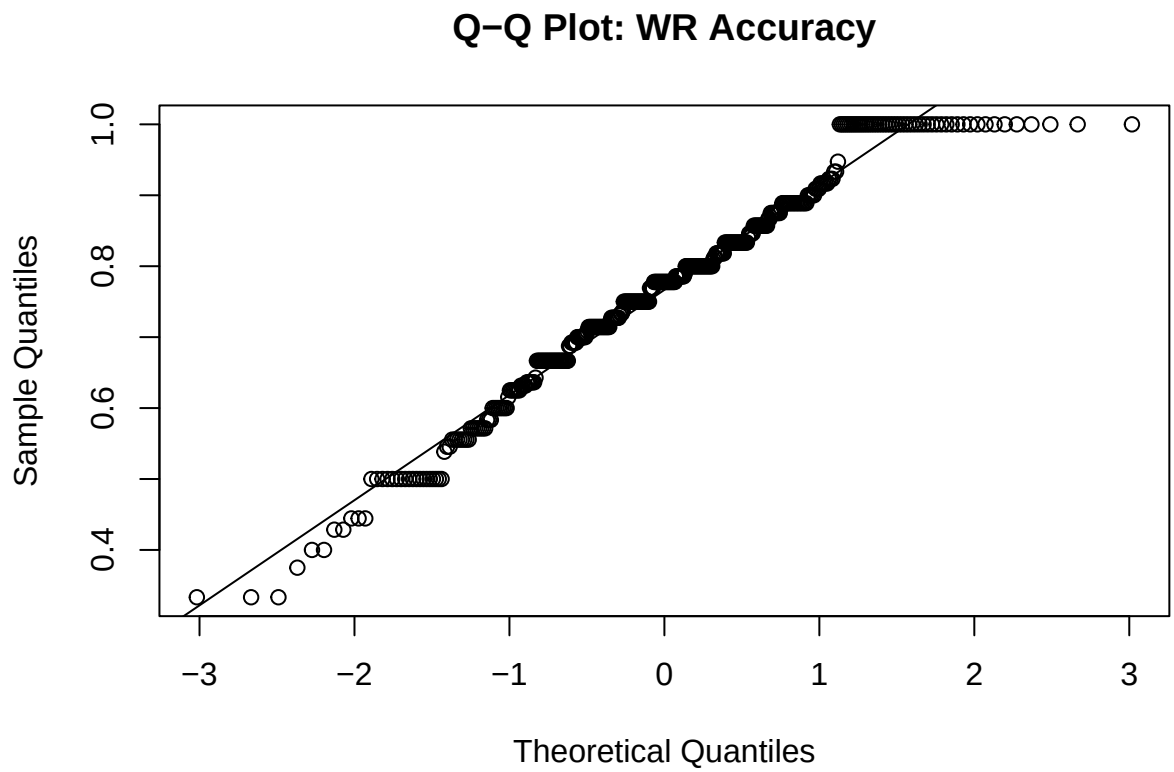
```r
raw_dataset <- read_csv(
  "sentence_memorability_data.csv",
  na = c("", "NA", "N/A"),
  show_col_types = FALSE
)

aggregated_matrix <- raw_dataset %>%
  filter(isTarget == TRUE) %>%
  arrange(participant_ID, Stimulus, Timestamp) %>%
  mutate(
    # mapping conditions
    Word_Condition = case_when(
  str_detect(Stimulus, "^HH_")  ~ "HH",
  str_detect(Stimulus, "^HVL_") ~ "HL",
  str_detect(Stimulus, "^LVH_") ~ "LH",
  str_detect(Stimulus, "^LVL_") ~ "LL"
),
    Word_Condition = factor(Word_Condition, levels = c("HH","HL","LH","LL")),

    Sentence_Voice = case_when(
      str_detect(Stimulus, "_A$") ~ "Active",
      str_detect(Stimulus, "_P$") ~ "Passive",
      TRUE ~ NA_character_
    ),
    Sentence_Voice = factor(Sentence_Voice, levels = c("Active", "Passive")),

    Accuracy.IR = suppressWarnings(as.numeric(Accuracy.IR)),
    Accuracy.WR = suppressWarnings(as.numeric(Accuracy.WR)),

    # store previous IR result
    prev_IR = lag(Accuracy.IR)
  ) %>%
  group_by(Stimulus, Word_Condition, Sentence_Voice) %>%
  summarise(

    # IR components
    Total_Repeats = sum(isRepeat == TRUE & Event == "Sentence shown", na.rm = TRUE),
    Total_Denom = sum(is.na(isRepeat) & Event == "Sentence shown"),
    Hits = sum(Event == "IR pressed" & isRepeat == TRUE & Accuracy.IR == 1, na.rm = TRUE),
    False_Alarms = sum(Event == "IR pressed" & is.na(isRepeat), na.rm = TRUE),

    # IR metrics
    Hit_Rate = if_else(Total_Repeats > 0, Hits / Total_Repeats, 0),
    False_Alarm_Rate = if_else(Total_Denom > 0, False_Alarms / Total_Denom, 0),
    IR_Memorability = Hit_Rate - False_Alarm_Rate,
```

```r
    # WR components (after correct IR)
    WR_correct = sum(Event == "WR pressed" & Accuracy.WR == 1 & prev_IR == 1, na.rm = TRUE),
    WR_total   = sum(Event == "WR pressed" & prev_IR == 1, na.rm = TRUE),

    # WR metric
    WR_Accuracy = if_else(WR_total > 0, WR_correct / WR_total, 0),

    .groups = "drop"
  ) %>%
  mutate(WR_Accuracy = replace_na(WR_Accuracy, 0))
```

```r
# Testing for Normality
# QQ Plots
qqnorm(aggregated_matrix$IR_Memorability, main = "Q-Q Plot: IR Memorability")
qqline(aggregated_matrix$IR_Memorability)
```




```r
qqnorm(aggregated_matrix$WR_Accuracy, main = "Q-Q Plot: WR Accuracy")
qqline(aggregated_matrix$WR_Accuracy)
```



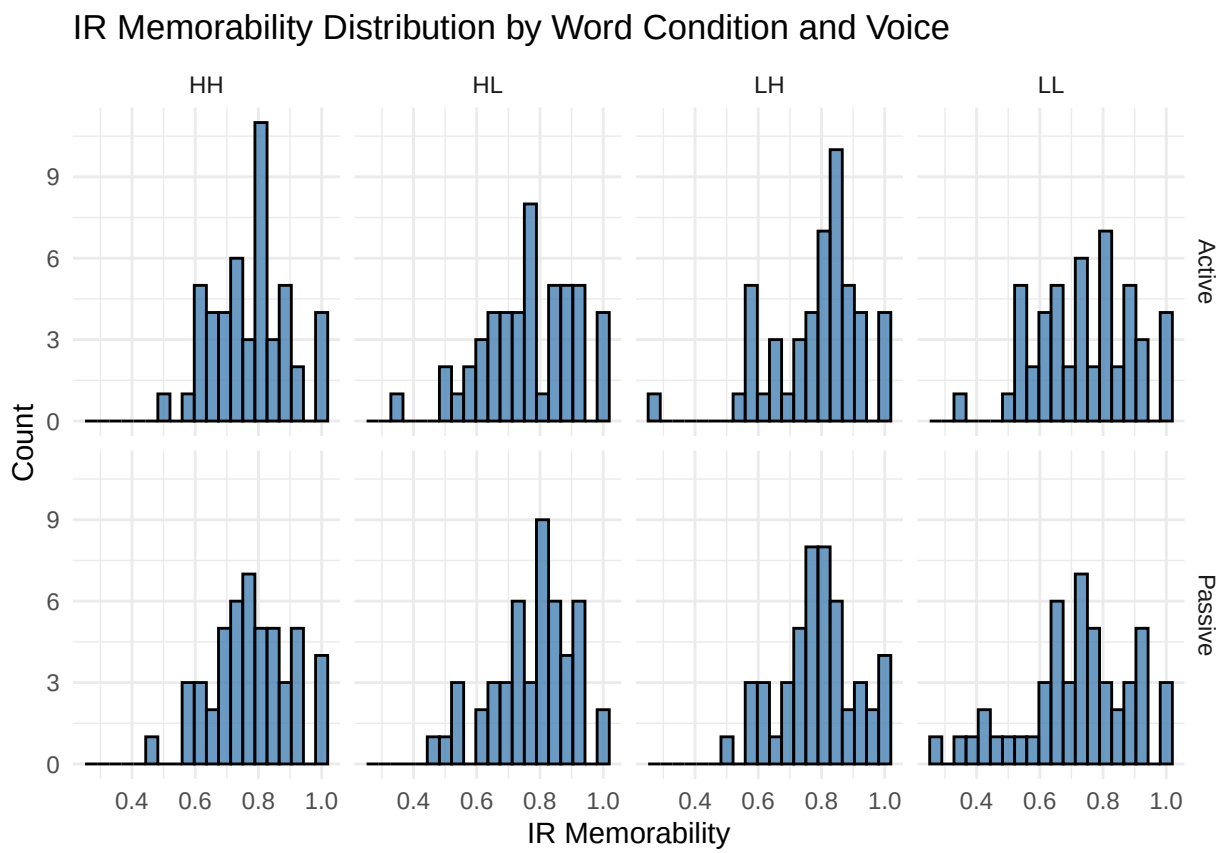

```r
# Shapiro Wilk
shapiro.test(aggregated_matrix$IR_Memorability)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  aggregated_matrix$IR_Memorability
## W = 0.97321, p-value = 1.256e-06
```

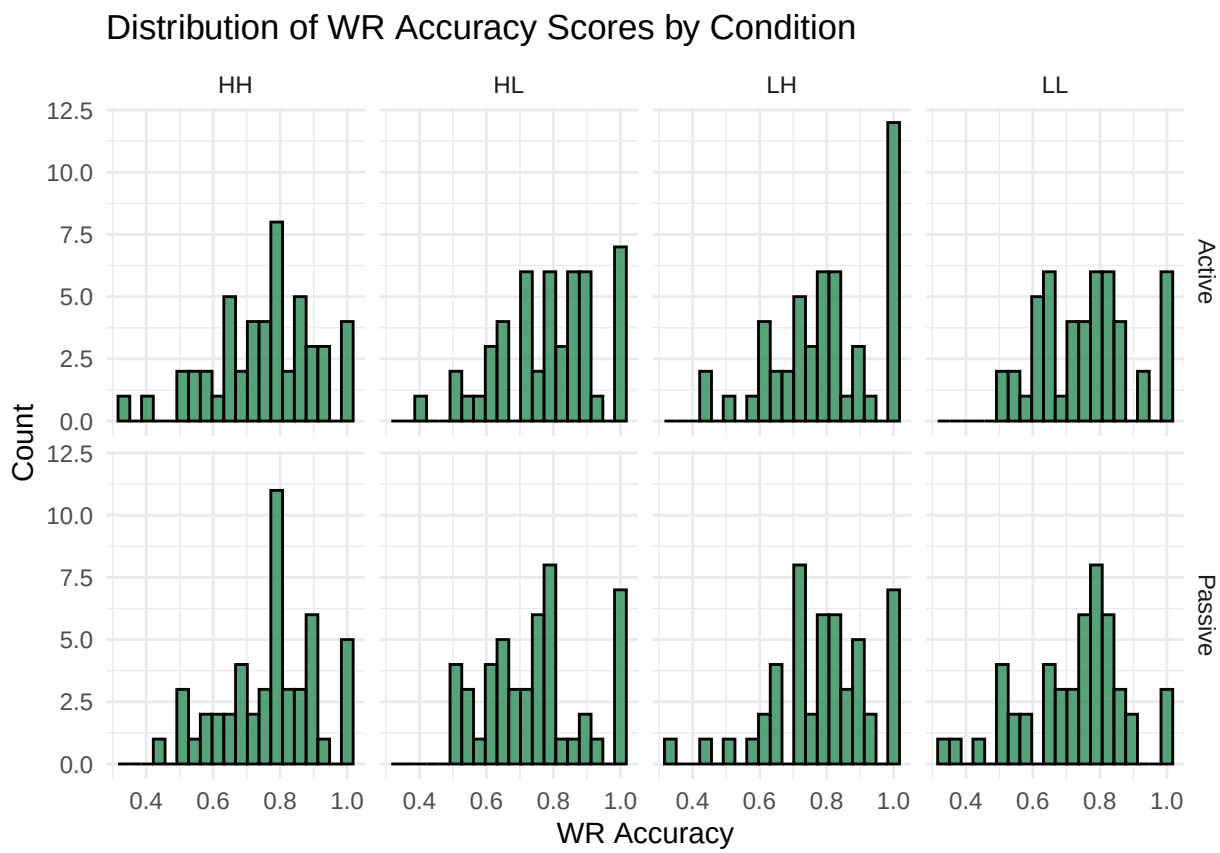
```r
shapiro.test(aggregated_matrix$WR_Accuracy)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  aggregated_matrix$WR_Accuracy
## W = 0.96837, p-value = 1.685e-07
```

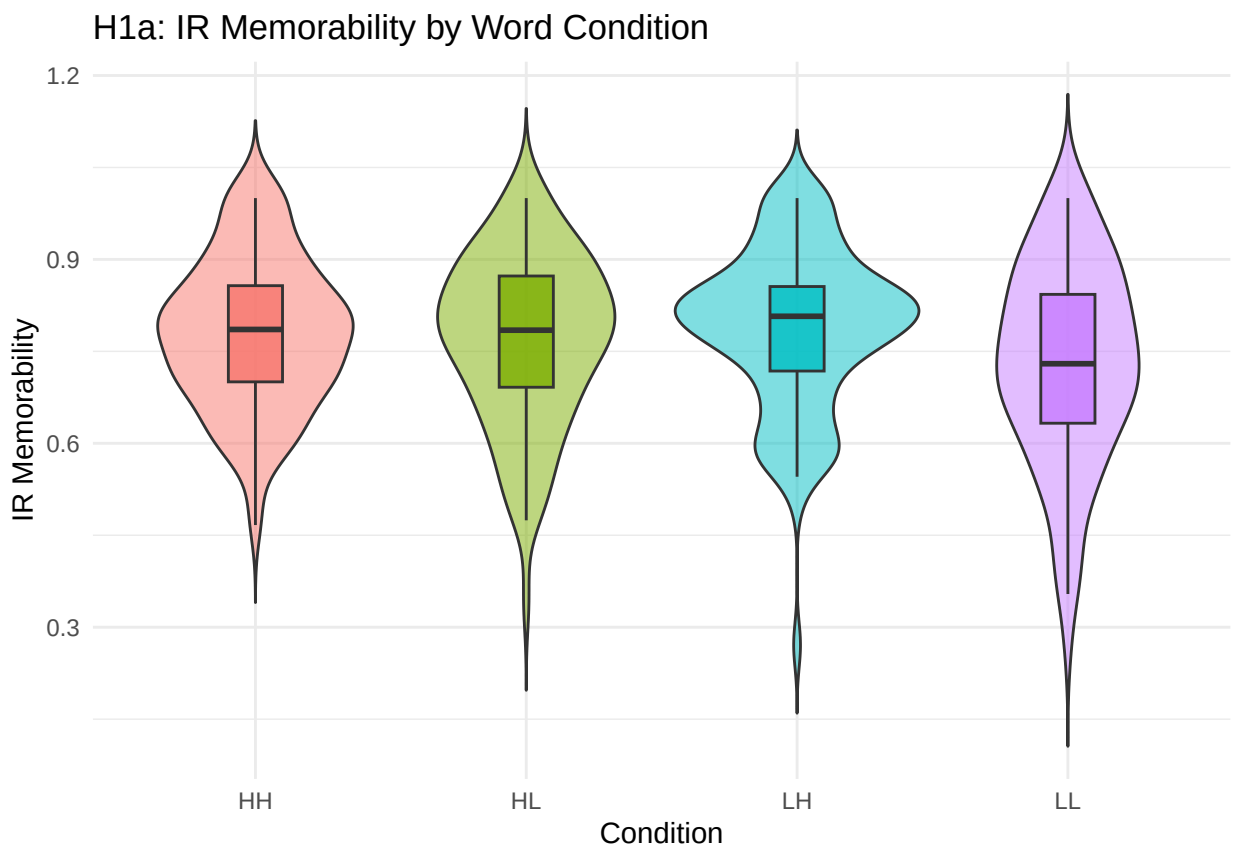
```r
# Graph 1: Histogram of IR Memorability Scores
ggplot(aggregated_matrix, aes(x = IR_Memorability)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "black", alpha = 0.8) +
  facet_grid(Sentence_Voice ~ Word_Condition) +
  labs(
    title = "IR Memorability Distribution by Word Condition and Voice",
    x = "IR Memorability",
    y = "Count"
  ) +
  theme_minimal()
```



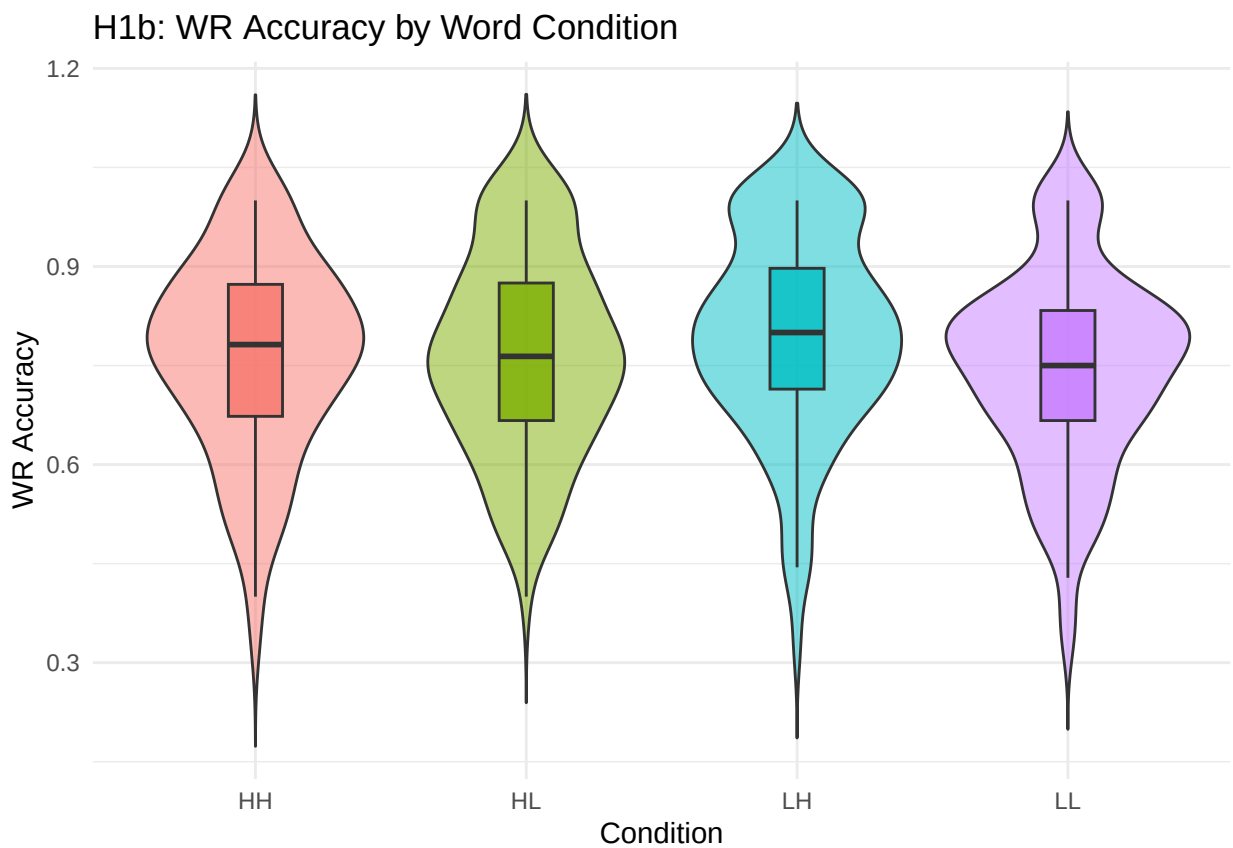

```r
# Graph 2: Histogram of WR Accuracy Scores
ggplot(aggregated_matrix, aes(x = WR_Accuracy)) +
  geom_histogram(bins = 20, fill = "seagreen", color = "black", alpha = 0.8) +
  facet_grid(Sentence_Voice ~ Word_Condition) +
  labs(
    title = "Distribution of WR Accuracy Scores by Condition",
    x = "WR Accuracy",
    y = "Count"
  ) +
  theme_minimal()
```



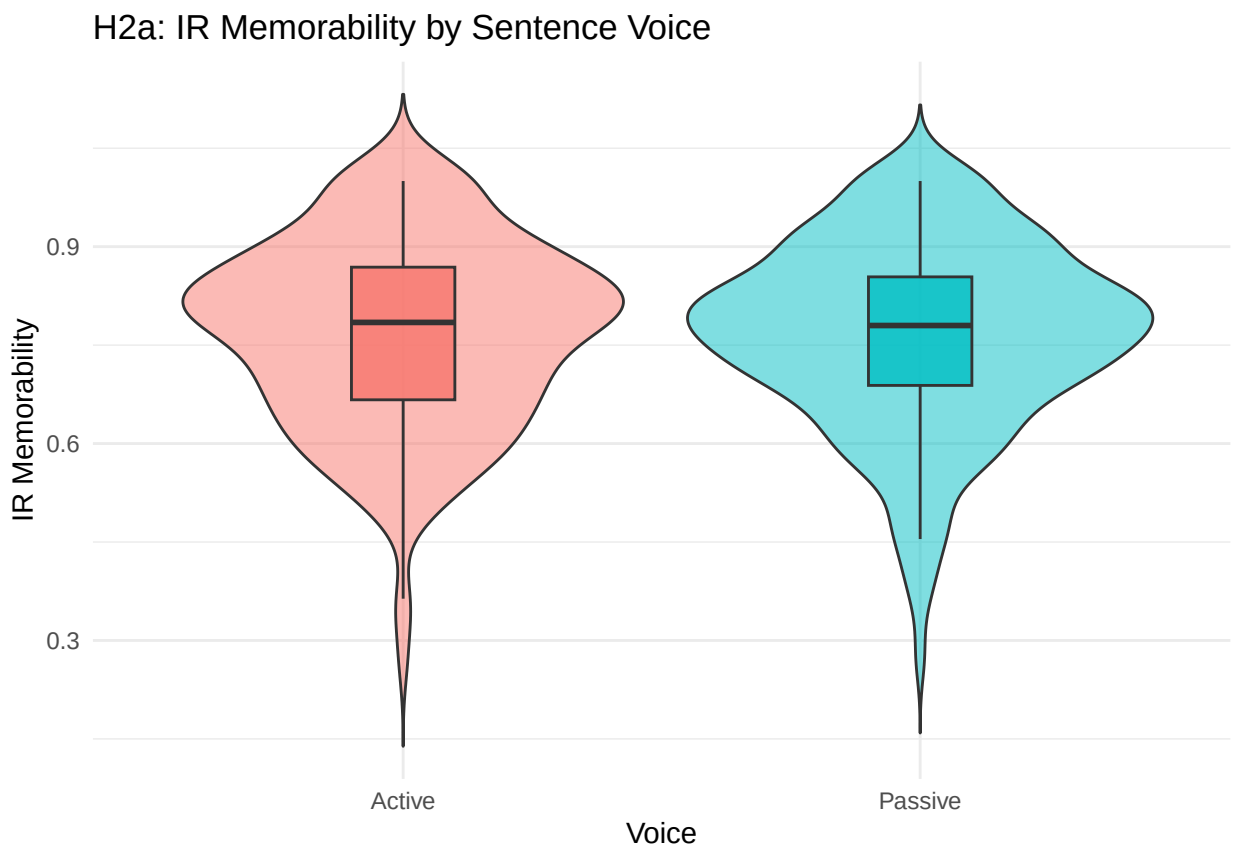

```r
# Graph 3 (H1a): Violin & Boxplot for IR by Word Condition
ggplot(aggregated_matrix, aes(x = Word_Condition, y = IR_Memorability, fill = Word_Condition)) +
  geom_violin(trim = FALSE, alpha = 0.5) +
  geom_boxplot(width = 0.2, outlier.shape = NA, alpha = 0.8) +
  labs(title = "H1a: IR Memorability by Word Condition", x = "Condition", y = "IR Memorability") +
  theme_minimal() + theme(legend.position = "none")
```



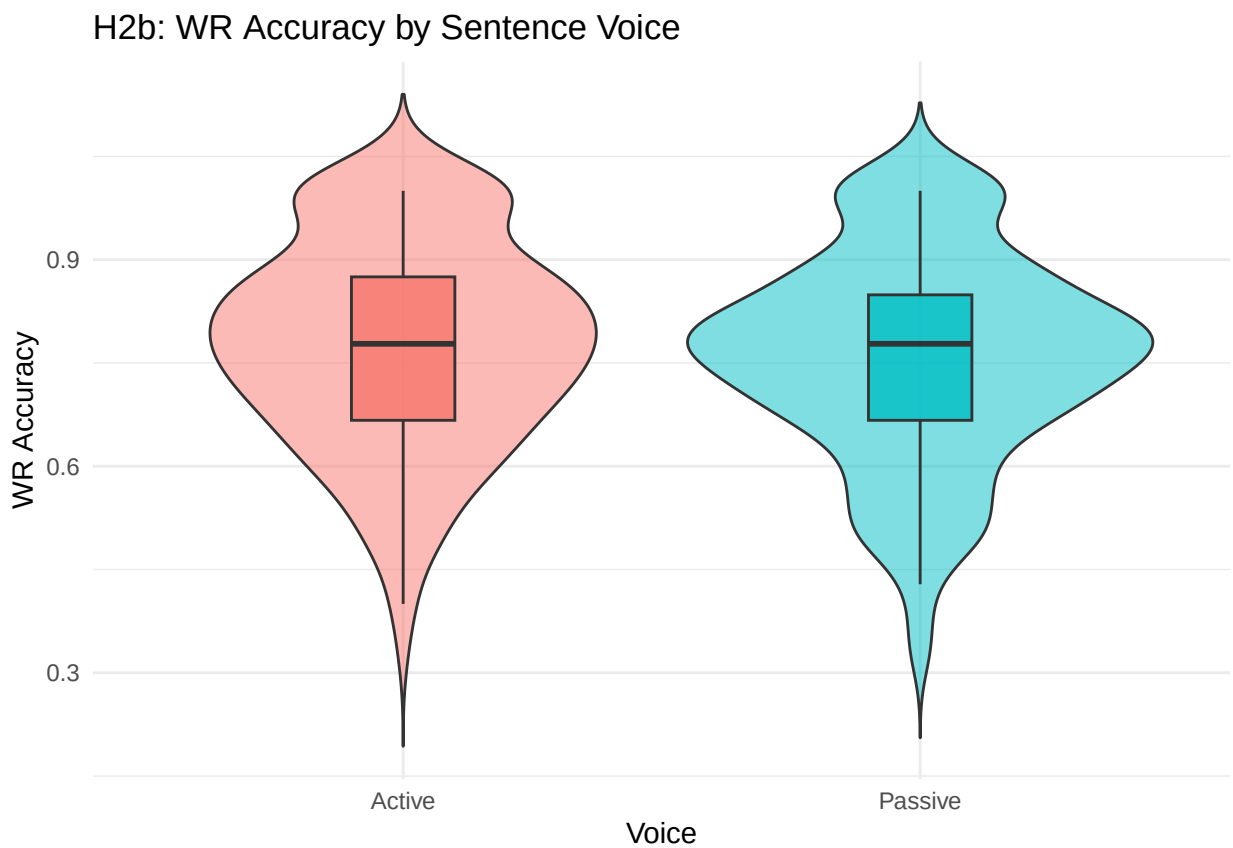

```r
# Graph 4 (H1b): Violin & Boxplot for WR by Word Condition
ggplot(aggregated_matrix, aes(x = Word_Condition, y = WR_Accuracy, fill = Word_Condition)) +
  geom_violin(trim = FALSE, alpha = 0.5) +
  geom_boxplot(width = 0.2, outlier.shape = NA, alpha = 0.8) +
  labs(title = "H1b: WR Accuracy by Word Condition", x = "Condition", y = "WR Accuracy") +
  theme_minimal() + theme(legend.position = "none")
```



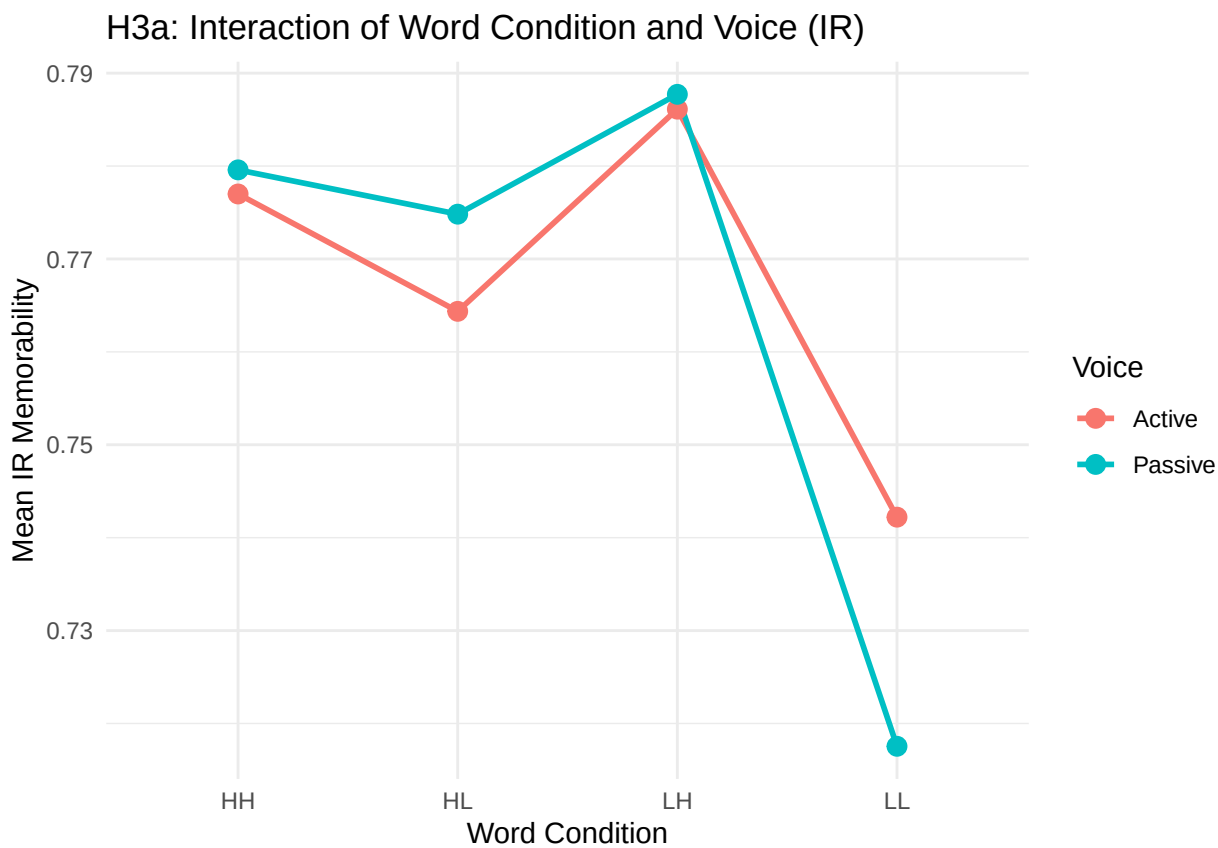

```
# Graph 5 (H2a): Violin & Boxplot for IR by Voice
ggplot(aggregated_matrix %>% drop_na(Sentence_Voice), aes(x = Sentence_Voice, y = IR_Memorability, fill
  geom_violin(trim = FALSE, alpha = 0.5) +
  geom_boxplot(width = 0.2, outlier.shape = NA, alpha = 0.8) +
  labs(title = "H2a: IR Memorability by Sentence Voice", x = "Voice", y = "IR Memorability") +
  theme_minimal() + theme(legend.position = "none")
```



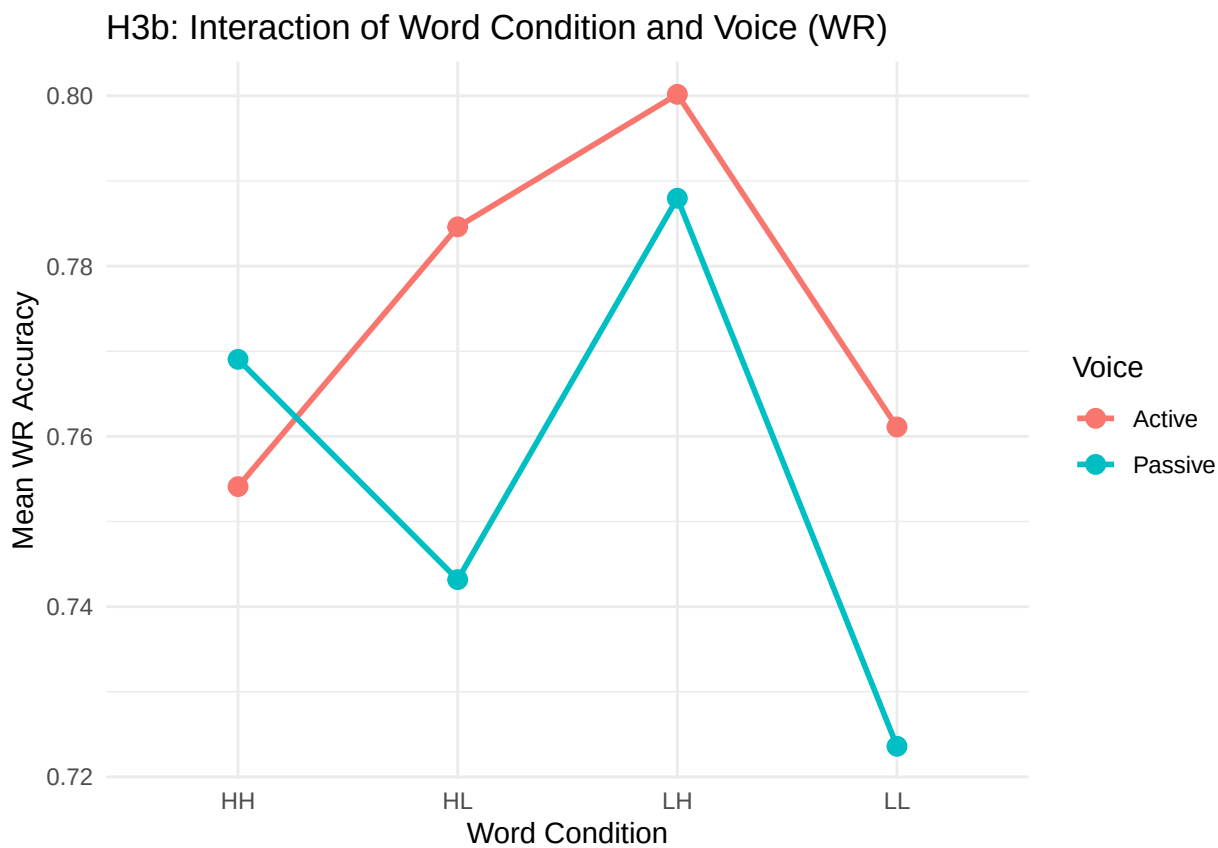

```
# Graph 6 (H2b): Violin & Boxplot for WR by Voice
ggplot(aggregated_matrix %>% drop_na(Sentence_Voice), aes(x = Sentence_Voice, y = WR_Accuracy, fill = S
  geom_violin(trim = FALSE, alpha = 0.5) +
  geom_boxplot(width = 0.2, outlier.shape = NA, alpha = 0.8) +
  labs(title = "H2b: WR Accuracy by Sentence Voice", x = "Voice", y = "WR Accuracy") +
  theme_minimal() + theme(legend.position = "none")
```




```
interaction_data <- aggregated_matrix %>%
  drop_na(Word_Condition, Sentence_Voice) %>%
  group_by(Word_Condition, Sentence_Voice) %>%
  summarise(
    Mean_IR = mean(IR_Memorability, na.rm = TRUE),
    Mean_WR = mean(WR_Accuracy, na.rm = TRUE),
    .groups = 'drop'
  )

# Graph 7 (H3a): Interaction Plot for IR
ggplot(interaction_data, aes(x = Word_Condition, y = Mean_IR, color = Sentence_Voice, group = Sentence_
  geom_line(linewidth = 1) + geom_point(size = 3) +
  labs(title = "H3a: Interaction of Word Condition and Voice (IR)", x = "Word Condition", y = "Mean IR
  theme_minimal()
```

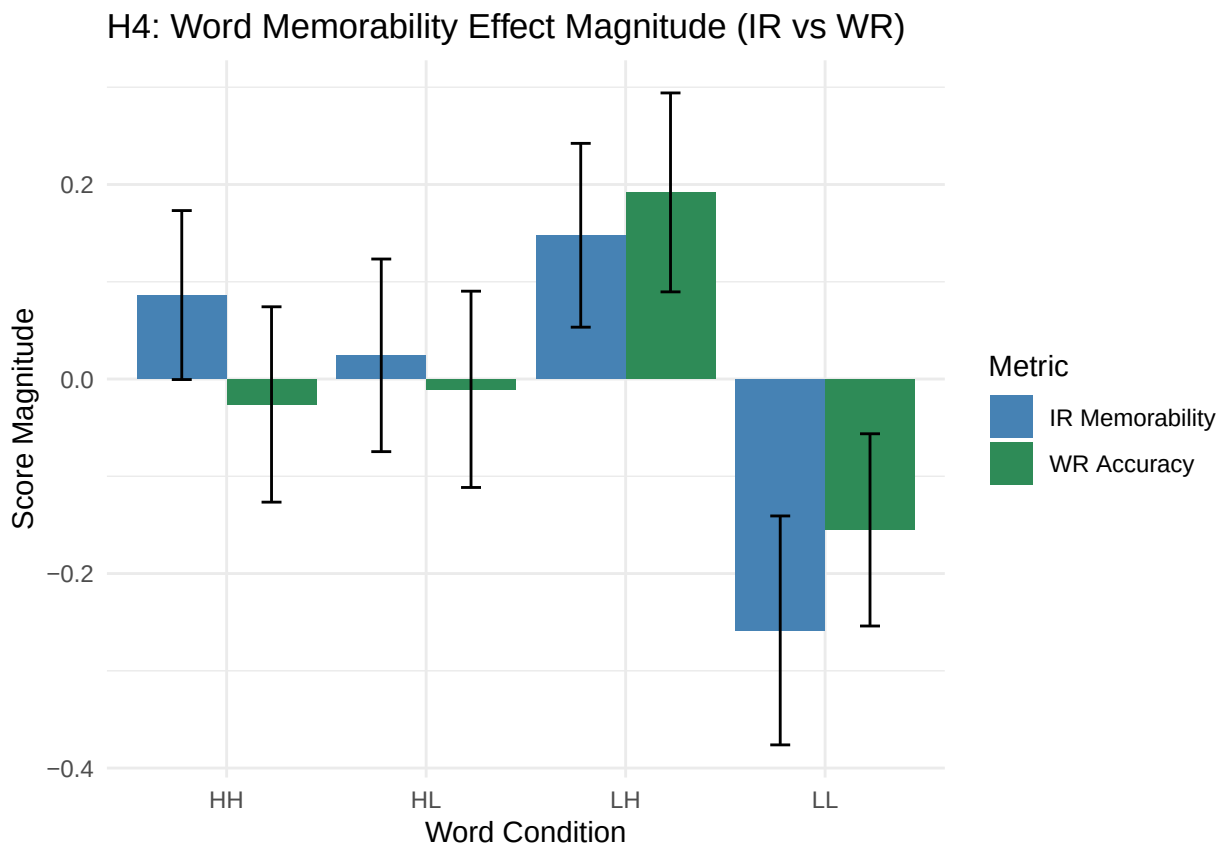



```
# Graph 8 (H3b): Interaction Plot for WR
ggplot(interaction_data, aes(x = Word_Condition, y = Mean_WR, color = Sentence_Voice, group = Sentence_V
  geom_line(linewidth = 1) + geom_point(size = 3) +
  labs(title = "H3b: Interaction of Word Condition and Voice (WR)", x = "Word Condition", y = "Mean WR A
  theme_minimal()
```

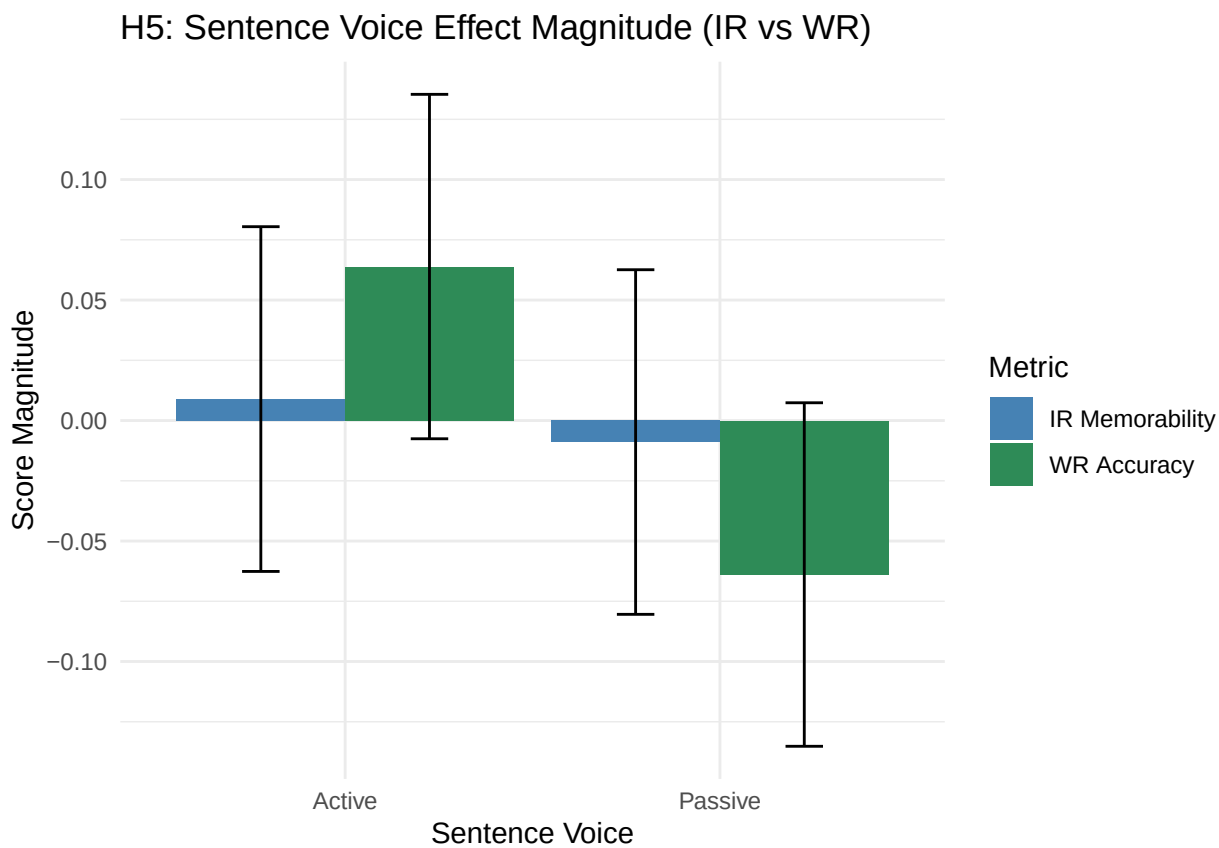



```r
aggregated_scaled <- aggregated_matrix %>%
  mutate(
    IR_scaled = as.numeric(scale(IR_Memorability)),
    WR_scaled = as.numeric(scale(WR_Accuracy))
  )

# Calculate average scaled IR and WR scores and their standard errors for each word condition
effect_data <- aggregated_scaled %>%
  drop_na(Word_Condition, Sentence_Voice) %>%
  group_by(Word_Condition) %>%
  summarise(
    IR_mean = mean(IR_scaled, na.rm=TRUE),
    IR_se = sd(IR_scaled, na.rm=TRUE)/sqrt(n()),
    WR_mean = mean(WR_scaled, na.rm=TRUE),
    WR_se = sd(WR_scaled, na.rm=TRUE)/sqrt(n()),
    .groups="drop"
  )

# Graph 9 (H4): Word Memorability effects on IR vs WR
effect_data_long <- effect_data %>%
  pivot_longer(cols = c(IR_mean, WR_mean), names_to = "Metric", values_to = "Mean") %>%
  mutate(SE = ifelse(Metric == "IR_mean", IR_se, WR_se))

ggplot(effect_data_long, aes(x = Word_Condition, y = Mean, fill = Metric)) +
  geom_bar(stat = "identity", position = position_dodge()) +
  geom_errorbar(aes(ymin = Mean - SE, ymax = Mean + SE), width = 0.2, position = position_dodge(0.9)) +
```

```
  labs(title = "H4: Word Memorability Effect Magnitude (IR vs WR)", x = "Word Condition", y = "Score Mag
  scale_fill_manual(values = c("steelblue", "seagreen"), labels = c("IR Memorability", "WR Accuracy")) +
  theme_minimal()
```




```
# Graph 10 (H5): Sentence Voice effects on IR vs WR
voice_effect <- aggregated_scaled %>%
  drop_na(Sentence_Voice) %>%
  group_by(Sentence_Voice) %>%
  summarise(
    IR_mean = mean(IR_scaled, na.rm=TRUE),
    IR_se = sd(IR_scaled, na.rm=TRUE)/sqrt(n()),
    WR_mean = mean(WR_scaled, na.rm=TRUE),
    WR_se = sd(WR_scaled, na.rm=TRUE)/sqrt(n()),
    .groups="drop"
  ) %>%
  pivot_longer(cols = c(IR_mean, WR_mean), names_to = "Metric", values_to = "Mean") %>%
  mutate(SE = ifelse(Metric == "IR_mean", IR_se, WR_se))

ggplot(voice_effect, aes(x = Sentence_Voice, y = Mean, fill = Metric)) +
  geom_bar(stat = "identity", position = position_dodge()) +
  geom_errorbar(aes(ymin = Mean - SE, ymax = Mean + SE), width = 0.2, position = position_dodge(0.9)) +
  labs(title = "H5: Sentence Voice Effect Magnitude (IR vs WR)", x = "Sentence Voice", y = "Score Magni
  scale_fill_manual(values = c("steelblue", "seagreen"), labels = c("IR Memorability", "WR Accuracy")) +
  theme_minimal()
```




```r
# Descriptive stats by Word Condition
desc_word <- aggregated_matrix %>%
  drop_na(Word_Condition) %>%
  group_by(Word_Condition) %>%
  summarise(
    IR_mean = mean(IR_Memorability, na.rm = TRUE),
    IR_sd = sd(IR_Memorability, na.rm = TRUE),
    IR_median = median(IR_Memorability, na.rm = TRUE),

    WR_mean = mean(WR_Accuracy, na.rm = TRUE),
    WR_sd = sd(WR_Accuracy, na.rm = TRUE),
    WR_median = median(WR_Accuracy, na.rm = TRUE),

    n = n(),
    .groups = "drop"
  )

print(desc_word)
```

```
## # A tibble: 4 x 8
##   Word_Condition IR_mean IR_sd IR_median WR_mean WR_sd WR_median     n
##   <fct>            <dbl> <dbl>     <dbl>   <dbl> <dbl>     <dbl> <int>
## 1 HH               0.778 0.121     0.786   0.762 0.148     0.782    98
## 2 HL               0.770 0.138     0.785   0.764 0.149     0.764    98
## 3 LH               0.787 0.131     0.807   0.794 0.151     0.8      98
## 4 LL               0.730 0.164     0.730   0.742 0.146     0.75     98
```

```r
# Descriptive stats by Sentence Voice
desc_voice <- aggregated_matrix %>%
  drop_na(Sentence_Voice) %>%
  group_by(Sentence_Voice) %>%
  summarise(
    IR_mean = mean(IR_Memorability, na.rm = TRUE),
    IR_sd = sd(IR_Memorability, na.rm = TRUE),
    IR_median = median(IR_Memorability, na.rm = TRUE),

    WR_mean = mean(WR_Accuracy, na.rm = TRUE),
    WR_sd = sd(WR_Accuracy, na.rm = TRUE),
    WR_median = median(WR_Accuracy, na.rm = TRUE),

    n = n(),
    .groups = "drop"
  )

print(desc_voice)
```

```
## # A tibble: 2 x 8
##   Sentence_Voice IR_mean IR_sd IR_median WR_mean WR_sd WR_median     n
##   <fct>            <dbl> <dbl>     <dbl>   <dbl> <dbl>     <dbl> <int>
## 1 Active           0.767 0.141     0.785   0.775 0.149     0.778   196
## 2 Passive          0.765 0.141     0.780   0.756 0.149     0.778   196
```

```r
kruskal_ir_word <- kruskal.test(IR_Memorability ~ Word_Condition, data = aggregated_matrix)
print(kruskal_ir_word)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  IR_Memorability by Word_Condition
## Kruskal-Wallis chi-squared = 7.3923, df = 3, p-value = 0.06039
```

```r
kruskal_wr_word <- kruskal.test(WR_Accuracy ~ Word_Condition, data = aggregated_matrix)
print(kruskal_wr_word)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  WR_Accuracy by Word_Condition
## Kruskal-Wallis chi-squared = 6.194, df = 3, p-value = 0.1025
```

```r
mw_ir_voice <- wilcox.test(
  IR_Memorability ~ Sentence_Voice,
  data = aggregated_matrix,
  exact = FALSE
)

print(mw_ir_voice)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  IR_Memorability by Sentence_Voice
## W = 19350, p-value = 0.8996
```

```
## alternative hypothesis: true location shift is not equal to 0
```

```r
mw_wr_voice <- wilcox.test(
  WR_Accuracy ~ Sentence_Voice,
  data = aggregated_matrix,
  exact = FALSE
)

print(mw_wr_voice)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  WR_Accuracy by Sentence_Voice
## W = 20505, p-value = 0.2468
## alternative hypothesis: true location shift is not equal to 0
```

```r
# Create categorical correctness variables
processed_trials <- raw_dataset %>%
  filter(isTarget == TRUE) %>%
  arrange(participant_ID, Stimulus, Timestamp) %>%
  group_by(participant_ID, Stimulus) %>%
  mutate(
    Word_Condition = case_when(
      str_detect(Stimulus, "^HH_")  ~ "HH",
      str_detect(Stimulus, "^HVL_") ~ "HL",
      str_detect(Stimulus, "^LVH_") ~ "LH",
      str_detect(Stimulus, "^LVL_") ~ "LL"
    ),
    Word_Condition = factor(Word_Condition, levels = c("HH", "HL", "LH", "LL")),

    Sentence_Voice = case_when(
      str_detect(Stimulus, "_A$") ~ "Active",
      str_detect(Stimulus, "_P$") ~ "Passive",
      TRUE ~ NA_character_
    ),
    Sentence_Voice = factor(Sentence_Voice, levels = c("Active", "Passive")),

    Accuracy.IR = suppressWarnings(as.numeric(Accuracy.IR)),
    Accuracy.WR = suppressWarnings(as.numeric(Accuracy.WR)),
    prev_IR = lag(Accuracy.IR)
  ) %>%
  ungroup()

ir_data <- processed_trials %>%
  filter(Event == "IR pressed") %>%
  mutate(
    IR_Result = case_when(
      isRepeat == TRUE & Accuracy.IR == 1 ~ "Correct",
      is.na(isRepeat) ~ "Incorrect",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(IR_Result))
```

```r
wr_data <- processed_trials %>%
  filter(Event == "WR pressed", prev_IR == 1) %>%
  mutate(
    WR_Result = case_when(
      Accuracy.WR == 1 ~ "Correct",
      Accuracy.WR == 0 ~ "Incorrect",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(WR_Result))



ir_word_table <- table(ir_data$Word_Condition, ir_data$IR_Result)
fisher_ir_word <- fisher.test(ir_word_table)
print(fisher_ir_word)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  ir_word_table
## p-value = 0.002997
## alternative hypothesis: two.sided
```

```r
wr_word_table <- table(wr_data$Word_Condition, wr_data$WR_Result)
fisher_wr_word <- fisher.test(wr_word_table)
print(fisher_wr_word)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  wr_word_table
## p-value = 0.325
## alternative hypothesis: two.sided
```

```r
ir_voice_table <- table(ir_data$Sentence_Voice, ir_data$IR_Result)
fisher_ir_voice <- fisher.test(ir_voice_table)
print(fisher_ir_voice)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  ir_voice_table
## p-value = 1
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.7347999 1.3411942
## sample estimates:
## odds ratio
##  0.9927989
```

```r
wr_voice_table <- table(wr_data$Sentence_Voice, wr_data$WR_Result)
fisher_wr_voice <- fisher.test(wr_voice_table)
print(fisher_wr_voice)
```

```
##
```

```
##  Fisher's Exact Test for Count Data
##
## data:  wr_voice_table
## p-value = 0.3423
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##  0.9257805 1.2542198
## sample estimates:
## odds ratio
##   1.077499
```

```r
# Create categorical correctness variables

ir_word_table <- table(ir_data$Word_Condition, ir_data$IR_Result)
chisq_ir_word <- chisq.test(ir_word_table)
print(chisq_ir_word)
```

```
##
##  Pearson's Chi-squared test
##
## data:  ir_word_table
## X-squared = 14.112, df = 3, p-value = 0.002756
```

```r
wr_word_table <- table(wr_data$Word_Condition, wr_data$WR_Result)
chisq_wr_word <- chisq.test(wr_word_table)
print(chisq_wr_word)
```

```
##
##  Pearson's Chi-squared test
##
## data:  wr_word_table
## X-squared = 3.4461, df = 3, p-value = 0.3278
```

```r
ir_voice_table <- table(ir_data$Sentence_Voice, ir_data$IR_Result)
chisq_ir_voice <- chisq.test(ir_voice_table)
print(chisq_ir_voice)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  ir_voice_table
## X-squared = 5.1175e-28, df = 1, p-value = 1
```

```r
wr_voice_table <- table(wr_data$Sentence_Voice, wr_data$WR_Result)
chisq_wr_voice <- chisq.test(wr_voice_table)
print(chisq_wr_voice)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  wr_voice_table
## X-squared = 0.89244, df = 1, p-value = 0.3448
```

```r
# library(tidyverse)
# library(ggcorrplot)
#
# library(dplyr)
```

```
# library(ggcorrplot)
#
# #  The complete numeric correlation data
# cor_data <- aggregated_matrix %>%
#   select(
#     IR_Memorability,
#     WR_Accuracy,
#     Hit_Rate,
#     False_Alarm_Rate,
#     Hits,
#     False_Alarms,
#     Total_Repeats,
#     Total_Denom,
#     WR_correct,
#     WR_total
#   )
#
# # Variable order
# var_order <- c(
#   "IR_Memorability",
#   "Hit_Rate",
#   "False_Alarm_Rate",
#   "Hits",
#   "False_Alarms",
#   "Total_Repeats",
#   "Total_Denom",
#   "WR_Accuracy",
#   "WR_correct",
#   "WR_total"
# )
#
# # Compute Spearman correlation matrix
# cor_matrix <- cor(cor_data, method = "spearman", use = "complete.obs")
#
# # Reorder matrix
# cor_matrix <- cor_matrix[var_order, var_order]
#
# # Reorder correlation matrix and visualize pairwise relationships between variables using a Spearman
# cor_matrix_plot <- cor_matrix[rev(rownames(cor_matrix)), ]
#
# ggcorrplot(
#   cor_matrix_plot,
#   method = "square",
#   type = "lower",
#   lab = TRUE,
#   lab_size = 3,
#   hc.order = FALSE,
#   colors = c("#d73027", "white", "#1a9850"),
#   title = "Spearman Correlation Matrix: All Variables",
#   ggtheme = theme_minimal()
# )
```

```
ir_wr_counts <- raw_dataset %>%
  arrange(participant_ID, Stimulus, Timestamp) %>%
```

```r
  group_by(participant_ID) %>%
  mutate(
    Accuracy.IR = suppressWarnings(as.numeric(Accuracy.IR)),
    Accuracy.WR = suppressWarnings(as.numeric(Accuracy.WR)),
    prev_IR = lag(Accuracy.IR)
  ) %>%
  ungroup() %>%
  summarise(
    IR_Total = sum(Event == "IR pressed", na.rm = TRUE),
    IR_Correct = sum(Event == "IR pressed" & Accuracy.IR == 1 & isRepeat==TRUE, na.rm = TRUE),
    IR_Incorrect = sum(Event == "IR pressed" & Accuracy.IR == 0, na.rm = TRUE),

    WR_Total = sum(Event == "WR pressed" & prev_IR == 1, na.rm = TRUE),
    WR_Correct = sum(Event == "WR pressed" & Accuracy.WR == 1 & prev_IR == 1, na.rm = TRUE),
    WR_Incorrect = sum(Event == "WR pressed" & Accuracy.WR == 0 & prev_IR == 1, na.rm = TRUE)
  )

print(ir_wr_counts)
```

```
## # A tibble: 1 x 6
##   IR_Total IR_Correct IR_Incorrect WR_Total WR_Correct WR_Incorrect
##      <int>      <int>        <int>    <int>      <int>        <int>
## 1     4508       4317          191     3872       2968          904
```

Contributions:

Bhavani: Data processing and formulae Nishita: Plotting Graphs and Debugging Tejasvini: Organizing the graphs and plotting graphs